

Question	Answer	Marks	Guidance
8(a)(i)	diode	B1	
8(a)(ii)	capacitor	B1	
8(b)(i)	peak $V_{IN} = 3.6 \text{ V}$	A1	Correct to at least 2 significant figures. AFC.
8(b)(ii)	period = 0.040 s	A1	Correct to at least 2 significant figures. AFC. Allow '0.04 s' given to 1 significant figure.
8(c)	<u>Any three points from:</u> - the rectification is half-wave - frequency (of input) = 25 Hz or angular frequency (of input) = 160 rad s^{-1} - r.m.s. input voltage = 2.5 V - time constant (of smoothing circuit) = 0.058 s - ripple / variation in (output) voltage = 1.6 V or 44%	B3	<u>List Rules apply.</u> All numerical values must be given with a unit and be correct to at least 2 significant figures. The symbols V_{IN} and V_{OUT} may be used without explanation. Otherwise, quantities must be identified by name. Ignore points that are simply descriptions of or direct read-offs from the graphs. Ignore working space. No ECF. (157 rad s^{-1}). (2.546 V). Allow 2.55 V. Do not allow 2.6 V (double rounding). Allow range 0.056 s to 0.059 s.